

2019 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs — Bouncing ball (v-t graph)

A tennis ball is released from rest beneath a motion sensor. The graph of vertical velocity v against time t shows the fall, the bounce, and the rise to a new maximum height. (a) (i) SHOW the initial acceleration, (ii) find the release height. (b) With mass 57.0 g, (i) find the magnitude of the change in momentum during the bounce, (ii) find the magnitude of the average force the ball exerts on the ground. (c) Sketch $a(t)$ from 0 to 1.18 s.

Worked solution

(a)(i) Initial acceleration — SHOW

From the straight-line section $0 \rightarrow 0.50$ s:

$$\Delta v = -4.9 - 0.0 = -4.9 \text{ m s}^{-1}, \Delta t = 0.50 \text{ s}$$

$$a = \Delta v / \Delta t = (-4.9) / (0.50) = -9.8 \text{ m s}^{-2}.$$

Therefore, $a \approx -9.8 \text{ m s}^{-2}$ (shown).

(a)(ii) Release height

Use $s = ut + \frac{1}{2}at^2$ with $u = 0$ and $a = -9.8 \text{ m s}^{-2}$ over the 0.50 s fall:

$$s = 0 + \frac{1}{2}(-9.8)(0.50)^2 = -1.23 \text{ m (downwards)}.$$

Magnitude of the height: 1.2 m.

(b)(i) Change in momentum (magnitude)

Just before and just after the bounce: 4.9 m s^{-1} down, 4.0 m s^{-1} up.

Magnitude of $\Delta p = m (v_{\text{after}} - v_{\text{before}}) = 0.0570 \times (4.0 - (-4.9))$
 $= 0.507 \text{ kg m s}^{-1}$.

Answer: 0.51 kg m s^{-1} .

(b)(ii) Average force during the bounce (magnitude)

Impulse $J = \Delta p = F \Delta t$, with contact time $\Delta t \approx 0.77 - 0.50 = 0.27 \text{ s}$.

$F = \Delta p / \Delta t = 0.507 / 0.27 = 1.88 \text{ N}$.

Answer: 1.9 N.

(c) Sketch of acceleration $a(t)$

- From 0 to 0.50 s: constant $a \approx -9.8 \text{ m s}^{-2}$ (gravity).
- From 0.50 to 0.77 s (during contact): large constant positive acceleration ($\approx +33 \text{ m s}^{-2}$).
- From 0.77 to 1.18 s: constant $a \approx -9.8 \text{ m s}^{-2}$ again as the ball rises.

Final answers

(a)(i) $a \approx -9.8 \text{ m s}^{-2}$ (shown)

(a)(ii) Release height $\approx 1.2 \text{ m}$

(b)(i) $|\Delta p| \approx 0.51 \text{ kg m s}^{-1}$

(b)(ii) $F \approx 1.9 \text{ N}$

(c) $a(t)$: $-g$, then large $+a$ during contact, then $-g$

Revision tips

- From a v - t graph, acceleration is the gradient and displacement is the area under the graph.
- Impulse (area under F - t) equals change in momentum, $J = \Delta p = F\Delta t$.
- During a bounce, the direction of velocity reverses, so Δp adds the magnitudes of down and up speeds.